



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22MA201 –ENGINEERING MATHEMATICS II

UNIT 5 – ANALYTIC FUNCTION

UNIT 5 & LM 5 – BILINEAR TRANSFORMATION

UNIT -5

BILINEAR TRANSFORMATION

5.1 Objective

In this unit in addition to Bilinear Transformations, Cross ratio, normal form of Bilinear Transformations will be discussed. Special emphasis will be given on conformal mapping, special bilinear transformation, the mapping, $W = z^n$, $W = z^2$ and the inverse mapping $w = z^{1/2}$. At the end extended complex plane will be discussed.

5.2 Introduction

Before going to discuss Bilinear Transformation, we shall define

Bilinear Transformation

Transformation or mapping, conformal transformation and linear transformation.

The equation
$$\left. \begin{aligned} u &= u(x, y) \\ v &= v(x, y) \end{aligned} \right\} \dots\dots\dots (1)$$
 define a transformation or mapping by means

of which a correspondence between uv plane and xy -plane can be established. If corresponding to each point of xy -plane, there is a unique point of uv -plane and conversely, then such a transformation is called **one-one transformation**. The corresponding points in the two planes are called images of each other. Hence, by means of equations (1), a region or curve of xy - plane is said to transform or mapped on or represented by the corresponding curve of the uv -plane.

The linear transformation : A transformation of the form $w = az + b$, is called a linear transformation, where a and b are complex constants.

5.3 Bilinear Transformation or Mobius transformation :

A transformation of the form $w = \frac{az + b}{cz + d} \dots\dots\dots (1)$ is called a Bilinear transformation

of linear fractional transformation, where a, b, c, d are complex constants.

Such type of transformation was first studied by Mobius and hence it is sometimes called mobius transformation.

Equation (1) may be derived as $CW_z + Wd - az - b = 0$ which is clearly a linear both in W and Z . That is why, it is called Bilinear. Here we assume that $ad - bc \neq 0$ which is called the determinant of the transformation. The transformation (1) is called **normalized** if $ad - bc = 1$ / if the determinant vanishes, then w is merely a constant as shown below.

Let W_1 and W_2 be two values of W corresponding to Z_1 and Z_2 in (1), then

$$\begin{aligned} W_2 - W_1 &= \frac{az_2 + b}{cz_2 + d} - \frac{az_1 + b}{cz_1 + d} \\ &= \frac{acz_1z_2 + bcz_1 + az_2d + bd - acz_1z_2 - bcz_2 - az_1d - bd}{(cz_1 + d)(cz_2 + d)} \\ &= \frac{z_2(ad - bc) - z_1(ad - bc)}{(cz_1 + d)(cz_2 + d)} \end{aligned}$$

$$= \frac{(z_2 - z_1)(ad - bc)}{(Cz_1 + d)(Cz_2 + d)} = 0 \text{ if } ad - bc = 0$$

$\therefore W_2 - W_1 = 0$ if $ad - bc = 0$ This shows that W is a constant

5.3.1 Critical Points : Let

$$w = \frac{az + b}{Cz + d} \quad \dots\dots\dots (1)$$

$$\therefore z = \frac{b - wd}{wc - a} \quad \dots\dots\dots (2)$$

$$\text{From (1) } \frac{dw}{dz} = \frac{ad - bc}{(cz + d)^2}$$

This means $\frac{dw}{dz} = \infty$ if $z = -d/c$

and $\frac{dw}{dz} = 0$ if $z = \infty$

The points $z = -d/c, z = \infty$ are called critical points where the conformal property does not hold good.

Also from (2) it is clear that for each $z \neq -d/c$ we have a value of W and for each $W \neq a/c$ there corresponds a value of Z and the correspondence between W and Z is one-one. The exceptional points $z = -d/c$ and $W = a/c$ are mapped into the points $W = \infty$ and $z = \infty$ respectively.

5.3.2 Extended complex plane : These points will not remain exceptional if we adjoin a new point called point at infinity denoted by ∞ to the complex plane and the complex plane in this case is called extended complex plane. Thus the critical point $z = \infty$ corresponds to the point $W = a/c$.

Hence we can now say that the bilinear sets up a one-one correspondence between the points of entire extended complex z -plane upon the entire extended w -plane.

5.4 Resultants or Product of two bilinear transformations

Consider two bilinear transformation

$$W = \frac{az + b}{cz + d} \text{ such that } ad - bc \neq 0 \quad \dots\dots\dots (1)$$

$$\text{and } \xi = \frac{a_1 w + d_1}{c_1 w + d_1} \text{ such that } a_1 d_1 - b_1 c_1 \neq 0 \quad \dots\dots\dots (2)$$

Putting the value of W from (1) in equation (2), we get

$$\xi = \frac{a_1 \left(\frac{az + b}{cz + d} \right) + d_1}{c_1 \left(\frac{az + b}{cz + d} \right) + d_1} = \frac{(aa_1 + b_1 c)z + (b_1 d + a_1 b)}{(c_1 a + d_1 c)z + (d_1 d + c_1 b)}$$

$$\text{Writing } A = aa_1 + b_1 c, \quad B = b_1 d + a_1 b, \quad C = c_1 a + d_1 c, \quad D = d_1 d + c_1 b$$

$$\text{We get } \xi = \frac{AZ + B}{CZ + D}$$

$$\begin{aligned} \text{Here } AD - BC &= (aa_1 + b_1 c)(d_1 d + c_1 b) - (b_1 d + a_1 b)(c_1 a + d_1 c) \\ &= aa_1 d_1 d + bb_1 cc_1 - b_1 c_1 ad - a_1 d_1 bc \\ &= (a_1 d_1 - b_1 c_1) (ad - bc) \neq 0 \text{ by (1) \& (2)} \end{aligned}$$

$$\text{Thus } \xi = \frac{AZ + B}{CZ + D} \text{ such that } AD - BC \neq 0$$

This is a bilinear transformation. This bilinear transformation is called the resultant or product of the transformation resultant (1) and (2). Thus we may say bilinear transformation forms a group.

5.4.1 Geometrical Inversion : Every bilinear transformation is the resultant of bilinear transformation with simple geometrical imports.

Consider the bilinear transformation

$$w = \frac{az + b}{cz + d} \quad \dots\dots\dots (1)$$

Where $ad - bc \neq 0, c \neq 0$

$$\text{from (1) } w = \frac{a}{c} \cdot \frac{z + \left(\frac{b}{a}\right)}{z + \left(\frac{d}{c}\right)} = \frac{a}{c} \left[\frac{1 + \frac{b/a - d/c}{z + d/c}}{1} \right]$$

$$w = \frac{a}{c} + \frac{bc - ad}{c^2} \cdot \frac{1}{z + d/c}$$

$$\text{Putting } Z_1 = z + d/c, Z_2 = \frac{1}{Z_1}, Z_3 = \frac{bc - ad}{c^2} \cdot Z_2$$

$$\text{We obtain } w = \frac{a}{c} + Z_3 \text{ which is similar to } Z_1 = z + d/c.$$

The above three auxiliary transformations namely Z_1, Z_2, Z_3 are of the form

$$W = Z + \alpha, W = \frac{1}{Z}, W = \beta Z$$

This proves that every general bilinear transformation is the resultant of the bilinear transformations

$$W = Z + \alpha, W = \frac{1}{Z}, W = \beta Z$$

Where (i) $W = Z + \alpha$ represents translation

(ii) $W = \frac{1}{Z}$ represents inversion

and (iii) $W = \beta Z$ represents dilation.

5.5 Cross Ratio

If Z_1, Z_2, Z_3, Z_4 are distinct points, then the ratio $\frac{(Z_4 - Z_1)(Z_2 - Z_3)}{(Z_2 - Z_1)(Z_4 - Z_3)}$ is called

the cross ratio of Z_1, Z_2, Z_3, Z_4 . This ratio is invariant under the bilinear transformation.

Number of four distinct cross-ratio :

From four points Z_1, Z_2, Z_3 and Z_4 lying in the Z -plane, we can obtain different cross ratios according to the order in which the points are taken. Since the four points can permute themselves in only six-distinct cross ratios. This is so

Bilinear Transformation

because if we interchange any points and then interchange remaining two, the cross ratios of the points in this new order will be the same. In fact, we have

$$(Z_1, Z_2, Z_3, Z_4) = (Z_2, Z_1, Z_4, Z_3) = (Z_3, Z_4, Z_1, Z_2) = (Z_4, Z_3, Z_2, Z_1) \dots\dots\dots (1)$$

$$(Z_1, Z_2, Z_4, Z_3) = (Z_2, Z_1, Z_3, Z_4) = (Z_3, Z_4, Z_2, Z_1) = (Z_4, Z_3, Z_1, Z_2) \dots\dots\dots (2)$$

$$(Z_1, Z_3, Z_2, Z_4) = (Z_2, Z_4, Z_1, Z_3) = (Z_3, Z_1, Z_4, Z_2) = (Z_4, Z_2, Z_3, Z_1) \dots\dots\dots (3)$$

$$(Z_1, Z_3, Z_4, Z_2) = (Z_2, Z_4, Z_3, Z_1) = (Z_3, Z_1, Z_2, Z_4) = (Z_4, Z_2, Z_1, Z_3) \dots\dots\dots (4)$$

$$(Z_1, Z_4, Z_2, Z_3) = (Z_2, Z_3, Z_1, Z_4) = (Z_3, Z_2, Z_4, Z_1) = (Z_4, Z_1, Z_3, Z_2) \dots\dots\dots (5)$$

$$(Z_1, Z_4, Z_3, Z_2) = (Z_2, Z_3, Z_4, Z_1) = (Z_3, Z_2, Z_1, Z_4) = (Z_4, Z_1, Z_2, Z_3) \dots\dots\dots (6)$$

5.5.1 The cross ratio (Z_1, Z_2, Z_3, Z_4) is real iff the four points Z_1, Z_2, Z_3, Z_4 lie on a circle or on a straight line.

Proof : By definition of cross ratio

$$(Z_1, Z_2, Z_3, Z_4) = \frac{(Z_4 - Z_1)(Z_2 - Z_3)}{(Z_4 - Z_3)(Z_2 - Z_1)} = \frac{(Z_4 - Z_1) \cancel{(Z_4 - Z_3)}}{(Z_2 - Z_1) \cancel{(Z_2 - Z_3)}}$$

$$\Rightarrow \arg \left(\frac{Z_4 - Z_1}{Z_4 - Z_3} \cdot \frac{Z_2 - Z_3}{Z_2 - Z_1} \right) = \arg \left(\frac{Z_4 - Z_1}{Z_4 - Z_3} \right) - \arg \left(\frac{Z_2 - Z_1}{Z_2 - Z_3} \right)$$

$= 0$ or $\pm \pi$ if points lie on a circle or straight line

5.6 Preservance of Cross Ratio :

Prove that cross ratio remains invariant under a bilinear transformation.

Proof : Let w_1, w_2, w_3, w_4 be the images of four distinct points z_1, z_2, z_3, z_4 respectively under a bilinear transformation.

$$w = \frac{az + b}{cz + d} \text{ where } ad - bc \neq 0 \dots\dots\dots (1)$$

If we show that

$$\frac{(w_4 - w_1)(w_2 - w_3)}{(w_2 - w_1)(w_4 - w_3)} = \frac{(z_4 - z_1)(z_2 - z_3)}{(z_2 - z_1)(z_4 - z_3)} \quad \dots\dots\dots (2)$$

The result will follow from (1)

$$\begin{aligned} W_r - W_s &= \frac{az_r + b}{cz_r + d} - \frac{az_s + b}{cz_s + d} \\ \Rightarrow W_r - W_s &= \frac{(ad - bc)(z_r - z_s)}{(cz_r + d)(cz_s + d)} \quad \dots\dots\dots (3) \end{aligned}$$

This $\Rightarrow (w_4 - w_1)(w_2 - w_3)$

$$\begin{aligned} &= \frac{(ad - bc)(z_4 - z_1)}{(cz_4 + d)(cz_1 + d)} \cdot \frac{(ad - bc)(z_2 - z_3)}{(cz_2 + d)(cz_3 + d)} \\ \Rightarrow (w_4 - w_1)(w_2 - w_3) &= \frac{(ad - bc)^2 (z_4 - z_1)(z_2 - z_3)}{(cz_1 + d)(cz_2 + d)(cz_3 + d)(cz_4 + d)} \end{aligned}$$

Similarly again from (3) prove that

$$(w_2 - w_1)(w_4 - w_3) = \frac{(ad - bc)^2 (z_2 - z_1)(z_4 - z_3)}{(cz_1 + d)(cz_2 + d)(cz_3 + d)(cz_4 + d)}$$

Dividing the last two results we get the required equation (2).

5.7 Fixed Point of a Bilinear Transformation :

To prove that in general there are two values of Z (invariant points) for which $w=z$, but there is only one if

$$(a - d)^2 + 4bc = 0$$

Show that if there are distinct invariant points p and q, the transformation may be put in the form (normal form)

$$\frac{w - p}{w - q} = \frac{k(z - p)}{(z - q)} \quad \text{and that if, there is only one invariant points p, the}$$

transformation may be put in the form (normal form) $\frac{1}{w - p} = \frac{1}{z - p} + k$.

Example 1 : Find the fixed point and the normal form of the following bilinear transformation and classify their nature.

$$(a) w = \frac{3iz + 1}{z + i} \quad (b) w = \frac{3z - 4}{z - 1}$$

Solution : (a) Putting $w = z$, then fixed points are given by

$$z = \frac{3iz + 1}{z + i} \Rightarrow z^2 + iz - 3iz - 1 = 0$$

$$\Rightarrow z^2 - 2iz - 1 = 0 \Rightarrow z^2 - 2iz + i^2 = 0$$

$$\Rightarrow (z - i)^2 = 0$$

$$\text{or, } z = i, i.$$

Hence there is only one distinct fixed points namely i .

$$\begin{aligned}\text{Now } \omega - i &= \frac{3iz + 1}{z + i} - i = \frac{3iz + 1 - iz - i^2}{z + i} \\ &= \frac{2iz + 1 + 1}{z + i} = \frac{2iz + 2}{z + i} \\ \Rightarrow \frac{1}{\omega - i} &= \frac{z + i}{2iz + 2} = \frac{z + i}{2i(z - i)} = \frac{1}{2i} \frac{z - i + 2i}{z - i} \\ &= \frac{1}{2i} \left(1 + \frac{2i}{z - i} \right) = \frac{1}{2i} + \frac{1}{z - i}\end{aligned}$$

or, $\frac{1}{\omega - i} = \frac{1}{z - i} - \frac{i}{2}$ which is the normal form. The transformation is parabola.

Solution: (b) $\omega = \frac{3z - 4}{z - 1}$

Putting $\omega = z$, $z = \frac{3z - 4}{z - 1}$

$$\Rightarrow z(z - 1) = 3z - 4 \Rightarrow z^2 - z - 3z + 4 = 0$$

$$\Rightarrow (z - 2)^2 = 0.$$

Thus $z = 2$ is the only fixed point so the transformation is parabolic. For

normal form $\omega - 2 = \frac{3z - 4}{z - 1} - 2 = \frac{z - 2}{z - 1}$

$$\therefore \frac{1}{\omega - 2} = \frac{z - 1}{z - 2} = \frac{z - 2 + 1}{z - 2} = \frac{1}{z - 2} + 1$$

Above is the required normal form.

Example 2. Find the bilinear transformation which maps the points $z_1 = 2$, $z_2 = i$ and $z_3 = -2$ into the points $\omega_1 = 1$, $\omega_2 = i$ and $\omega_3 = -1$.

Solution : The required transformation is given by

$$\frac{(\omega - \omega_1)(\omega_2 - \omega_3)}{(\omega_1 - \omega_2)(\omega_3 - \omega)} = \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)} \quad \dots (1)$$

Substituting the given values as above (1) becomes

Bilinear Transformation

$$\frac{(\omega - 1)(i + 1)}{(1 - i)(-1 - \omega)} = \frac{(z - 2)(i + z)}{(z - 0)(-2 - z)}$$

$$\Rightarrow -\frac{(\omega - 1)(1 + i)(1 + i)}{(\omega + 1)(1 - i)(1 + i)} = -\frac{(z - 2)(2 + i)(2 + i)}{(z + 2)(2 - i)(2 + i)}$$

$$\Rightarrow \frac{(\omega - 1)(1 + i)^2}{(\omega + 1)[1 - i^2]} = \frac{(z - 2)(2 + i)^2}{(z + 2)(2^2 - i^2)}$$

$$\Rightarrow \frac{(\omega - 1).2i}{(\omega + 1).2} = \frac{(z - 2)(3 + 4i)}{(z + 2).5}$$

$$\Rightarrow \frac{\omega - 1}{\omega + 1} = \frac{(z - 2)(3 + 4i)}{(z + 2)5i}$$

$$\Rightarrow \frac{(\omega - 1)}{(\omega + 1)} = \frac{(z - 2)i(4 - 3i)}{5(z + 2)i}$$

$$\Rightarrow \frac{\omega - 1}{\omega + 1} = \frac{(z - 2)(4 - 3i)}{5(z + 2)}$$

Using componendo & dividendo, we get

$$\frac{\omega - 1 + (\omega + 1)}{(\omega + 1) - (\omega - 1)} = \frac{(z - 2)(4 - 3i) + 5(z + 2)}{5(z + 2) - (z - 2)(4 - 3i)}$$

$$\Rightarrow \frac{2\omega}{2} = \frac{4z - 8 - 3iz + 6i + 5z + 10}{5z + 10 - 4z + 8 + 3iz - 6i}$$

$$\Rightarrow \omega = \frac{9z - 3iz + 2 + 6i}{z + 3iz + 18 - 6i} = \frac{3(3 - i)z + 2(1 + 3i)}{z(1 + 3i) + 6(3 - i)}$$

$$= \frac{3z + 2 \frac{(1 + 3i)}{3 - i}}{\frac{(1 + 3i)}{3 - i}z + 6} = \frac{3z + 2i}{iz + 6}$$

which is required transformations.

Example 3. Find the bilinear transformation which carries 0, i , $-i$ and 1, -1 , 0.

Solution : Consider z_1, z_2, z_3 as $0, i, -i$ and $\omega_1, \omega_2, \omega_3$ as $1, -1, 0$. By cross ratio we have

$$\begin{aligned} \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)} &= \frac{(\omega - \omega_1)(\omega_2 - \omega_3)}{(\omega_1 - \omega_2)(\omega_3 - \omega)} \\ \Rightarrow \frac{(z - 0)(i - (-i))}{(0 - i)(-i - z)} &= \frac{(\omega - 1)(-1 - 0)}{(1 - (-1))(0 - \omega)} \\ \Rightarrow \frac{2iz}{i(i + z)} &= \frac{-(\omega - 1)}{-2\omega} \\ \Rightarrow 4 \omega z &= (\omega - 1)(z + i) \\ \Rightarrow 4 \omega z &= (\omega - 1)z + (\omega + 1)i \\ \Rightarrow 4 \omega z - \omega z + z &= \omega i - i \\ \Rightarrow \omega (4z - z - i) &= -z - i \\ \Rightarrow \omega (3z - i) &= -z - i \\ \Rightarrow \omega &= \frac{-(z + i)}{-(-3z + i)} = \frac{z + i}{-3z + i} \quad \dots (1) \end{aligned}$$

Which is of the form $\omega = f(z) = \frac{az + b}{cz + d}$, at $ad - bc \neq 0$

The relation (1) is called bilinear transformation.

Example 4. Show that the transformation $\omega = \frac{2z + 3}{z - 4}$ transforms the circle $x^2 + y^2 - 4x = 0$ into the straight line $4u + 3 = 0$ and explain the curve obtained is not a circle.

Solution : We have $\omega = \frac{2z + 3}{z - 4}$ or $(z - 4)\omega = 2z + 3$

$$\Rightarrow \omega z - 4\omega = 2z + 3 \Rightarrow z(\omega - 2) = 3 + 4\omega$$

$$\Rightarrow z = \frac{3 + 4\bar{\omega}}{\omega - 2}$$

The equation of the circle in z -plane is

$$\begin{aligned} x^2 + y^2 - 4x &= 0 \\ \Rightarrow z \bar{z} - 2(z + \bar{z}) &= 0 \end{aligned}$$

Bilinear Transformation

By virtue of the given transformation this is transformed to

$$\frac{(3 + 4\omega)(3 + 4\bar{\omega})}{(3 + 4\omega)(3 + 4\bar{\omega})} = 0$$

$$(\omega - 2)(\bar{\omega} - 2) - 2 \frac{M}{N} + \frac{P}{Q} = 0$$

$$\text{or, } 9 + 16 \omega \bar{\omega} + 12 (\omega + \bar{\omega}) - 2[(3 + 4\omega)(\bar{\omega} - 2) + (3 + 4\bar{\omega})(\omega - 2)] = 0$$

$$\text{or, } 9 + 16 \omega \bar{\omega} + 12 (\omega + \bar{\omega}) - 2 [(6 + 4\omega\bar{\omega} - 3\bar{\omega} - 8\omega)(4\omega\bar{\omega} + 3\omega - 6 - 8\bar{\omega})] = 0$$

$$\text{or, } 22 (\omega + \bar{\omega}) + 33 = 0$$

$$\text{or, } 2 (\omega + \bar{\omega}) + 3 = 0$$

or, $4u + 3 = 0$ which is the eqn. of a straight line in ω -plane.

Hence curve thus obtained is not a circle.

DISCUSSION QUESTIONS:

1. What do you mean by critical points?
2. What are the possible images of a circle under a linear transformation?
3. What are the two important properties of Bilinear Transformation?
4. Bilinear Transformation always transforms circles into -----